

LUCA MARZARI

Postdoctoral Research Fellow
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🌐 <https://lmarza.github.io>

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EDUCATION

Ph.D. in Computer Science (cum laude)

October 2022 - March 2026

University of Verona

📍 Verona, Italy

Dissertation title: "Advanced Neural Networks Verification for Safe and Explainable Intelligent Systems"

My thesis proposed novel, efficient, and reliable methods for verifying and enhancing the safety and explainability of deep neural networks, particularly in the context of deep reinforcement learning applications. These methods have been applied in simulation and realistic domains, ranging from ground and aquatic robots to surgery assistants and games.

Advisors: Prof. Alessandro Farinelli, Ph.D., and Prof. Ferdinando Cicalese, Ph.D.

Examination committee: Prof. Ezio Bartocci, Ph.D. and Prof. Nils Jansen

Master's Degree in Computer Eng. (110/110 cum laude)

October 2020 - July 2022

University of Verona

📍 Verona, Italy

Dissertation title: "Deep Reinforcement Learning for Robotic Tasks: Enhancing Efficiency and Safety"

My master's thesis proposed two innovative techniques based on a hierarchical approach, either by decomposing a complex task into simpler subtasks for a robotic pick-and-place task or by providing the agent with incremental knowledge about the environment for mapless navigation tasks. The results of this work were published in the proceedings of the IEEE International Conference on Advanced Robotics and the ACM/SIGAPP Symposium on Applied Computing (SAC IRMAS 2022).

EXPERIENCE

Vienna University of Technology (TU Wien) - Postdoctoral Research Associate

April 2026-

RobustifAI EU project: Robustifying Generative AI through human-centric integration of neural and symbolic methods.

📍 Vienna, AT

Advisor: Prof. Ezio Bartocci, Ph.D.

Imperial College London - Visiting Researcher

Feb 2026-March 2026

Research visit at the Centre for Explainable AI. Invitation: Prof. Francesco Leofante, Ph.D.

📍 London, UK

University of Verona - Research Associate

Oct 2025 - March 2026

Researcher on the BEHAVE- Learning Safe Behaviours for human-robot cooperation project.

📍 Verona, IT

Advisor: Prof. Alessandro Farinelli, Ph.D.

Carnegie Mellon University - Visiting Researcher

Jul 2023 - Feb 2024

Research visit at the Intelligent Control Lab for the development of *ModelVerification.jl* verification tool.

📍 Pittsburgh (PA), USA

Advisor: Prof. Changliu Liu, Ph.D.

PUBLICATIONS

Up-to-date research indicators are available in my [Google Scholar profile](#). (* indicates equal contribution when not first-author)

Journal papers:

- [J.1] Probabilistically Robust Counterfactual Explanations under Model Changes
Artificial Intelligence Journal (AIJ), Q1 2025
L. Marzari, F. Leofante, F. Cicalese and A. Farinelli
- [J.2] Probabilistically Tightened Linear Relaxation-based Perturbation Analysis for Neural Network Verification
Journal of Artificial Intelligence Research (JAIR), Q1 2025
L. Marzari, F. Cicalese and A. Farinelli
- [J.3] Verifying Online Safety Properties for Safe Deep Reinforcement Learning
ACM Transaction on Intelligent Systems and Technologies (TIST), Q1 2025
L. Marzari, F. Cicalese, A. Farinelli, C. Amato and E. Marchesini
- [J.4] Designing Control Barrier Function via Probabilistic Enumeration for Safe Reinforcement Learning Navigation
IEEE Robotics and Automation Letters (RA-L), Q1 2025
L. Marzari, F. Trotti, E. Marchesini and A. Farinelli
- [J.5] ϵ -Retraining Reinforcement Learning Algorithms
Journal of Autonomous Agents and Multi-Agent Systems (JAAMAS) Q2 2026
L. Marzari, C. Liu, P.L. Donti, and E. Marchesini

Conference papers:

- [C.1] A Survey on the Verification of Reinforcement Learning Policies
International Joint Conference on Artificial Intelligence (IJCAI) 2026
L. Marzari, E. Bartocci and E. Marchesini

- [C.2] [Probabilistic Verification of Recurrent Neural Networks for Single and Multi-Agent Reinforcement Learning](#)
International Joint Conference on Artificial Intelligence (IJCAI) 2026
L. Marzari and E. Marchesini
- [C.3] [On the Probabilistic Learnability of Compact Neural Network Preimage Bounds](#)
AAAI Conference on Artificial Intelligence (AAAI) 2026
L. Marzari, M. Bicego, F. Cicalese., and A. Farinelli
- [C.4] [Enumerating Safe Regions in Deep Neural Networks with Provable Probabilistic Guarantees](#)
AAAI Conference on Artificial Intelligence (AAAI) 2024
L. Marzari, D. Corsi, E. Marchesini, A. Farinelli and F. Cicalese
- [C.5] [The #DNN-Verification Problem: Counting Unsafe Inputs for Deep Neural Networks](#)
International Joint Conference on Artificial Intelligence (IJCAI) 2023
L. Marzari, D. Corsi, F. Cicalese, and A. Farinelli
- [C.6] [Improving Policy Optimization via \$\epsilon\$ -Retrain](#)
International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2025
L. Marzari, P.L. Donti, C.Liu, and E. Marchesini
- [C.7] [Rigorous Probabilistic Guarantees for Robust Counterfactual Explanations](#)
European Conference on Artificial Intelligence (ECAI) 2024
L. Marzari, F. Leofante, F. Cicalese, and A. Farinelli
- [C.8] [Online Safety Property Collection and Refinement for Safe Deep Reinforcement Learning in Mapless Navigation](#)
International Conference on Robotics and Automation (ICRA) 2023
L. Marzari, E. Marchesini, and A. Farinelli
- [C.9] [Advancing Neural Network Verification through Hierarchical Safety Abstract Interpretation](#)
European Conference on Artificial Intelligence (ECAI) 2025
L. Marzari, I. Mastroeni, and A. Farinelli
- [C.10] [Safe Deep Reinforcement Learning by Verifying Task-Level Properties](#)
International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2023
E. Marchesini*, **L. Marzari***, A. Farinelli, and C. Amato
- [C.11] [Constrained Reinforcement Learning and Formal Verification for Safe Colonoscopy Navigation](#)
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2024
D. Corsi*, **L. Marzari***, A. Pore*, A. Farinelli, A. Casals, P. Fiorini and D. Dall'Alba
- [C.12] [ModelVerification.jl: A Comprehensive Toolbox for Formally Verifying Deep Neural Networks](#)
International Conference on Computer-Aided Verification (CAV) 2025
T. Wei, H. Hu*, **L. Marzari***, K. Yun*, P. Niu*, X. Luo, and C.Liu
- [C.13] [RobustX: Robust Counterfactual Explanations Made Easy.](#)
International Joint Conference on Artificial Intelligence (IJCAI) 2025
J. Jiang*, **L. Marzari***, A. Purohit, and F. Leofante
- [C.14] [Verifying Learning-Based Robotic Navigation Systems.](#)
International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS) 2023
G. Amir, D. Corsi, R. Yerushalmi, **L. Marzari**, D. Harel, A. Farinelli, and G. Katz
- [C.15] [Curriculum Learning for Safe Mapless Navigation](#)
ACM/SIGAPP Symposium on Applied Computing (SAC IRMAS) 2022
L. Marzari, D. Corsi, E. Marchesini and A. Farinelli
- [C.16] [Towards Hierarchical Task Decomposition using Deep Reinforcement Learning for Pick and Place Subtasks](#)
IEEE International Conference on Advanced Robotics (ICAR) 2021
L. Marzari, A. Pore, D. Dall'Alba, G. Aragon-Camarasa, A. Farinelli and P. Fiorini

Workshop/Bridge papers (peer-reviewed)

- [W.1] [Formal Verification for Counting Unsafe Inputs in Deep Neural Networks](#)
2nd Workshop on Formal Verification of Machine Learning - International Conference on Machine Learning (ICML) 2023
L. Marzari, D. Corsi, F. Cicalese, and A. Farinelli
- [W.2] [Scaling #DNN-Verification Tools with Efficient Bound Propagation and Parallel Computing](#)
10th Italian Workshop on Artificial Intelligence and Robotics (AIRO) 2023
L. Marzari, G. Roncolato and A. Farinelli
- [W.3] [Tackling Environment Sustainability Challenges via Reinforcement Learning and Counterfactual Explanations](#)
Bridge Programme on Explainable AI, Energy and Critical Infrastructure Systems - AAAI Conference on Artificial Intelligence (AAAI) 2025
L. Marzari and E. Marchesini
- [W.4] [Explaining Reinforcement Learning Policies for Power Grid Operations](#)
CINI National Lab AIIS Conference on Artificial Intelligence (Ital IA) 2025
L. Marzari, F. Leofante and E. Marchesini

Conference and journal papers under review:

- [C.17] [Formal Verification of Variational Quantum Circuits](#)
N. Assolini*, **L. Marzari***, I. Mastroeni and A. Di Piero
- [C.18] [MIRA: Quantifying Neural Network Monitorability via Feature Space Analysis](#)
W. Essbai, L. Di Stefano, **L. Marzari** and E. Bartocci

Summary

Reference	Articles	Citations	h-index
Google Scholar	25	404	12
Scopus	24	206	8

FELLOWSHIPS, AWARDS, GRANTS AND RESEARCH PROJECTS

- **Project title:** PROVE-IT
Funding entity: Austrian Science Fund (FWF)
Principal Investigator: Luca Marzari
Funds: $\approx$ 350,000 EUR
Page: <https://www.fwf.ac.at/en/research-radar/10.55776/ESP1944725>
- **Project title:** "BEHAVE - Learning Safe Behaviours for human-robot cooperation"
Funding entity: Italian Ministry of University and Research
Principal Investigator: Alessandro Farinelli
Role: Proposal Writing author and Researcher (first-author papers: [J.2],[J.3], [J.4], [C.3],[C.9])
Funds: $\approx$ 200,000 EUR
Page: <https://isla-lab.github.io/projects/behave/>
- **Project title:** "ModelVerification.jl: a Comprehensive Toolbox for Formally Verifying Deep Neural Networks."
Funding entity: The Boeing Company and Carnegie Mellon University
Principal Investigator: Changliu Liu
Role: Toolbox documentation, tutorial, and main author of the final paper report (accepted at CAV 2025 [C.12]).
Page: <https://github.com/intelligent-control-lab/ModelVerification.jl>
- **Project title:** "RobustX: Robust Counterfactual Explanations Made Easy."
Funding entity: Imperial College London
Role: Toolbox developer, tutorial, and main author of the final paper report (accepted at IJCAI 2025 [C.13]).
Principal Investigator: Francesco Leofante
Page: <https://github.com/RobustCounterfactualX/RobustX>
- **3-years Doctoral Grant** for a PhD in Computer Science at University of Verona.
Funding entity: Italian Ministry of University and Research
Role: PhD Student Fellow
Funds: $\approx$ 41,000 EUR
- **Mobility Grant for non-EU destinations** for the research visit at Carnegie Mellon University (CMU), Pittsburgh (PA), USA.
Duration: 7 months
Funds: $\approx$ 3,400 USD
- **Travel Grants** Several awards for attending and presenting papers at international conferences such as the 40th AAAI Conference on Artificial Intelligence, 27th and 28th European Conference on Artificial Intelligence and the 24th International Conference on Autonomous Agents and Multiagent Systems
Funds: $\approx$ 3,600 USD

SELECTED SERVICE AND PROGRAM ORGANIZATION

- **Organizer and presenter** of the outreach activity *Rigorous Probabilistic Guarantees for Robust Counterfactual Explanations* at the 27th European Conference on Artificial Intelligence (ECAI 2024).
- **Program Committee member** for FMCAD 2026, IJCAI 2026, AAAI 2026-2027, AE TACAS 2026, XAISEC 2026.
- **Reviewer** for several international conferences and journal letters such as AAAI, ICML, NeurIPS, ICLR, IJCAI, ECAI, ICRA, AAMAS, RA-L, JAIR, EURO Journal on Computational Optimization.
- **Presenter** at several top-level international conferences such as IJCAI, AAAI, CAV, ICML, ICRA, AAMAS.
- Participation as an **exhibitor** at *Sea Drone tech summit 2022* to present the European Project INTCATCH 2020.

SELECTED PRESENTATIONS AND INVITED TALKS

- *"From Classical to Scalable Verification of Neural Networks: Challenges and Opportunities in Reinforcement Learning"*.
Host: EU RobustifAI Workshop IJCAI 2026 (link)
- *"Introduction to Deep Learning"*. Mini course of 8 hours.
Host Companies: Cromodora Wheels SpA and Salvagnini Group, Verona, Italy 2025
- *"Artificial Intelligence and Industry 5.0"*.
Host: Digital Transformation Training, SMACT Competence Center, Cortina, Italy 2025

- *"Enhancing safety of intelligent agents via Formal Verification and Safe Deep Reinforcement Learning"*.
Host: Prof. Christopher Amato, Northeastern University, Boston, USA 2024
- *"Formal and Probabilistic Verification for Deep Neural Networks"*.
Host: Prof. Changliu Liu, Carnegie Mellon University, Pittsburgh, USA 2024

SELECTED TEACHING AND MENTORING

Teaching Assistant or Instructor at the University of Verona from 2022 to 2026:

- *C Programming* (120 hours), first year of BSc in Computer Science
- *Artificial Intelligence* (30 hours), third year of BSc in Computer Science
- *Reinforcement Learning* (24 hours), first year of MSc in Artificial Intelligence
- *Planning and Automated Reasoning* (24 hours), first year of MSc in Artificial Intelligence
- *Programming and Database* (24 hours), first year of MSc in Data Science

Students Mentored

- Tosca Pierro, graduate student at the MSc in Computer Engineering and Automation at the Marche Polytechnic University, Italy 2026 (Visiting student at TU Wien).
Thesis: *"Enhancing Scalability of Verified Neural Hamilton-Jacobi Value Functions through Novel Verification-Guided Training"*.
- Buzzoni Riccardo, undergraduate student at the BSc in Computer Science at the University of Verona, Italy 2026.
Thesis: *"Benchmarking reinforcement learning algorithms for Power Grid Operations"*
- Pullia Domenico, undergraduate student at the BSc in Computer Science at the University of Verona, Italy 2026.
Thesis: *"Benchmarking reinforcement learning algorithms for Power Grid Operations"*
- Nicolò Squarzonì, graduate student at the MSc in Computer Science and Engineering at the University of Verona, Italy 2024.
Thesis: *"A Reinforcement Learning approach to Strategy Optimization for Battery Energy Storage Systems"*
- Gabriele Roncolato, graduate student at the MSc in Computer Science and Engineering at the University of Verona, Italy 2023.
Thesis: *"Accelerating Formal Verification Techniques for Neural Networks"*.
Part of the thesis has led to the production of the article [W.2]

REFERENCES

- Ezio Bartocci - Full Professor at Vienna University of Technology (TU Wien) (ezio.bartocci@tuwien.ac.at)
- Changliu Liu - Associate Professor at the Carnegie Mellon University (cliu6@andrew.cmu.edu)
- Enrico Marchesini - Postdoctoral Associate at MIT (incoming Assistant Professor) (emarche@mit.edu)
- Francesco Leofante - Assistant Professor at Imperial College London (f.leofante@imperial.ac.uk)
- Alessandro Farinelli - Full Professor at the University of Verona (alessandro.farinelli@univr.it)

I authorize the processing of my personal data contained in the curriculum vitae in accordance with Legislative Decree 30 June 2003, no. 196, and the GDPR (EU Regulation 2016/679).

Vienna, August 24, 2026
Luca Marzari